requires the concentration.

STAGES OF COGNITIVE PROCESSES

1. **Sensation:** It refers to people's awareness about different stimuli in the environment, which we experience in various modalities such as **vision, hearing, touch smell, and taste**.
2. **Attention:** Various sensory organs respond to the stimuli, which occur in the environment. Visual attention is based on the location and colour of the stimuli and auditory attention is based on pitch, timbre, intensity, etc.
3. **Perception:** It is the course of processing the information and clearing the meaning of stimuli that occurs around us. For example, when we see a pen on the table, we recognize it as an object used for writing.

Differences between Sensation and Perception:

* *The sensation* is the passive process of getting information from the environment to the body and brain. Passive process means, aquatically we can sense the stimuli, we do not have to consciously engage in the sensing process.
* *Perception* is the process that helps in choosing. organizing, and understanding of given data that is brought to the brain by using senses.

Working nature:

a. *Sensation occurs:* a) Sensory organs will receive the information from a physical stimulus in the environment. b) Sensory receptors will transmit this information to the neural impulses and then the brain.

b. *Perception follows* a) The brain makes this information more meaningful by organizing and comprehending process.

4. **Learning:** It is the process of gaining new knowledge and skills via experience and training. This acquired knowledge brings a change in the behaviour and adjustments in different settings of the environment. For example, we learn new languages, riding a car, and applying logical skills to solve many problems.
5. **Memory:** The information we process and learn is **registered and stored** in the memory system. Memory also helps us to easily **recover the stored information** when it is required for use. For example, writing the answers in the examination after studying for the exam.
6. **Thinking**: It is the process of using stored knowledge to perform various tasks. The logical relationship is established among various objects in our mind and a reasonable decision is taken to solve the problems.
7. **Decision-making:** It is the process of selecting an intellectual choice from various options. During this process, people try to select the option by considering the advantages and disadvantages of all the options available. Effective decision-making is done based on the forecast of the outcome of each option given, and based on all these logical calculations, the best option will be chosen.
8. **Problem-solving:** It is a mental process, which determines, evaluates, analyses the problem systematically to find the solution.

The main objective of problem-solving is to minimize the obstacles and issues to be resolved systematically.

The steps of problem-solving are as follows:

1. Problem identification.
2. Problem to be defined.
3. Formulation of strategies.
4. The obtained information to be organized.
5. Resources to be allocated as per the requirement.
6. Progress to be monitored continuously.
7. Results need to be evaluated.

6.1. ATTENTION- DEFINITION, TYPES, DETERMINANTS, DURATION, DEGREE, AND ALTERATION IN ATTENTION

INTRODUCTION

Attention is the term applied during the perceptual processes that select certain inputs for inclusion in our conscious experience, or awareness at any given time. It is the process that involves the act of listening and concentrating on a topic, object, or event for the attainment of desired ends.

Attention is the primary characteristic of the conscious mind that is required to acquire knowledge. Attention is also regarded as **the power of the mind,** which can be focussed on any object and which makes the person to be trained in a particular field of interest. Attention exists only when someone is attending to something.

Attention is the mental activity of the person experiencing a particular mode of knowledge and skills that he undergoes. Attention does not mean the simple awareness or consciousness of the object, person, or idea.

Many times we are very conscious of various things that happen around us, but we may attend to only one or two objects at one point in time.

For example:

When a student is listening to a class in a classroom, he may be aware of various noises outside the classroom.

While a nurse is calculating the dose of medicine for patients, she may be aware or conscious of so many stimuli that may be in the form of noise, smell and things, but the nurse is concentrating only on the dose of medicine that is to be administered to the patients.

From these examples, we conclude that attention is the highest consciousness of an object or stimulus out of many happenings. It means the focusing of consciousness on a particular object or idea at a particular time to an exclusion of all other objects or ideas.

Meaning and Definitions of Attention:

* Attention is an important factor responsible for gaining knowledge.
* It is an act of directing one's thought" towards a particular activity or object".
* Concentration or focusing of consciousness upon one object. Eg. Military command – prepares a soldier for an action –Woodworth.
* Attention is the behavioural and cognitive process of selectively concentrating on a discrete stimulus, while ignoring other perceivable stimuli.
* "Attention is the concentration of consciousness upon one object other than upon another" - ***Dumvllle (1938)***
* "Attention is the process of getting an object or thought clearly before the mind". - ***Ross (1951)***
* The process which compels the individual to select some particular stimulus according to his interest and attitude out of the multiplicity of stimuli present in the environment -Sharma R. N. (1967).
* Attention is the cognitive process of selectively concentrating on one aspect of the environment while ignoring other things. - ***John R. Anderson***
* Attention can be thought of as the bridge over which some parts of the external world the aspects selectively focussed on are brought into the subjective world of ours consciousness, so that we may regulate our behaviour. - ***Carver, and Schuler***

Characteristics of Attention:

* It is one of the mental activities, in which individuals react to specific incoming stimuli.
* Attention shifts from one object to another. It shows interest in new things. It **never focuses only on a single object**.
* Attention brings clarity to our consciousness on the object, which we attend.
* Attention is responsible for establishing an individual's attentive attitude.
* In the attention process, people use their sensory organs to advance and improve their response to incoming stimuli.

Types of Attention:

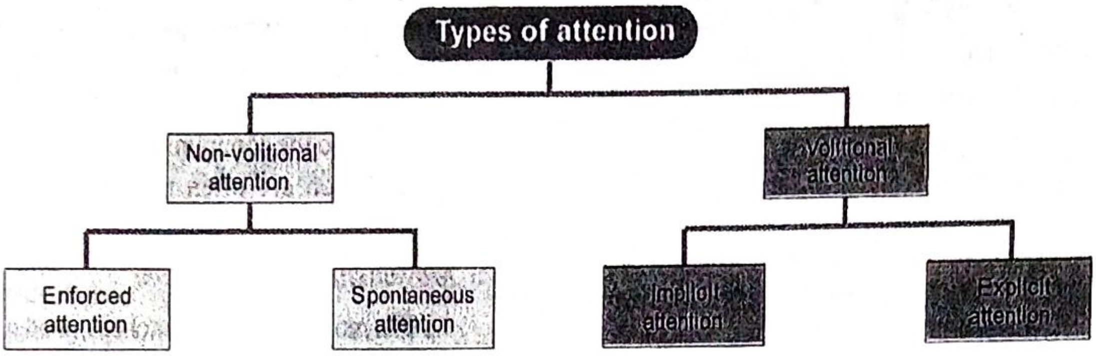

Fig. 6.1

Classification of Attention: Attention can be classified into mainly two types,

1. Non-Volitional (Involuntary attention) and
2. Volitional (Voluntary attention).

1. **Non-Volitional (Involuntary Attention):** This type of attention is stimulated without **cognizant efforts.** In this, we attend the objects **without imparting conscious efforts**. For example, Mother attends to her crying child automatically, our attention is more towards bright colours, beautiful nature, etc.

   They are two types;

   a. ***Enforced attention*** is sustained by instincts. Due to the striking qualities of stimulus, the individual is forced to attend or concentrate on it. For example, the attraction towords the opposite sex.

   b. ***Spontaneous attention*** is involuntary attention and it is produced by sentiments and it involves personal interest. For example, Mother's attention towards the child cry.

2. **Volitional (Voluntary Attention):** This involves **the will and demands the conscious effort to achieve certain goals.** There is always a goal for every act of attention to the object.

   They are two types;

   a. ***Implicit attention*** is obtained by a **single act of will**. For example, homework given by the teacher and the fear of punishment arouse attention in the student to complete the given tasks.

   b. ***Explicit attention*** is obtained by **repeated acts of will.** It requires strong willpower and strong motives. One has to do handwork to be attentive all the time. For example, Attention paid during the examination.

Types of attention based on the needs or circumstances:

1. ***Sustained Attention:***
   * In this attention, people **concentrate on objects consciously with certain objectives** for a long period by avoiding the sources of distractions or deviations.
   * It is the ability to pay attention to only one task consciously by concentrating on that particular task for a long time and by avoiding all other forms of distractions or deviations. ***For example:*** Be in meditation, reading a book, memorizing a chapter, or following a classroom lecture without distraction.
2. ***Selective Attention:*** In this attention, the people choose to pay attention only **to particular objects.** For example, nurses pay attention to CPR training.
3. ***Divided Attention:*** In divided attention, the people pay **attention to two or more tasks at the same time** and is also sometimes regarded as Multi-tasking, which involves juggling between two or more tasks at the same time. ***For example,*** texting somebody while attending a meeting. Divided attention uses mental focus on a very large scale; hence because, of divided attention, the user may get exhausted very quickly.
4. ***Alternating Attention:*** This attention is closely related to the divided attention, but is relatively different also, in the case of divided attention, we split our attention between two tasks, while in the case of alternating attention, **the entire attention is shifted from one task to another or is done alternately**. ***For example,*** nurses administer the medications after a sponge bath of a patient.

5. **Visual Attention:** The sensory organ such as the eye plays a vital role in imparting attention towards certain objects. Visual attention is essential while reading and watching visual objects. ***For example,*** nurses impart visual attention during the physical examination of the patients.

6. **Auditory Attention:**

* In this form of attention, people pay attention only to the sense of hearing. Paying attention to an important announcement can be an example of auditory attention.
* Auditory and visual attention both function in conjunction with each other. ***For example,*** nurses listen to complaints of their patients.

7. **Transient attention:** It is a short-term response to a stimulus that temporarily attracts or distracts attention.

Determinants of attention:

Determinants of attention are two types;

I. External factors or condition.

II. Internal factors.

I. **External Factors or Condition:** These factors are the features of outside stimuli, which make the strongest aid for catching the attention.

Types of external stimuli;

1. *Nature of the stimulus:*
   * All kinds of stimuli cannot capture the same degree of attention. The **pictorial representation captures the individual attention** easily than words. Handsome men or beautiful women are more attracted and capture the attention of others. ***For example,*** Video-assisted teaching captures the student's attention easier than a one-way lecture teaching method.
   * Thus, individual needs to choose the effective stimulus for capturing maximum attention.
2. *Intensity and size of the stimulus:*
   * A massive stimulus attracts more attention to the individual than a weak stimulus.
   * Our attention is easily captured towards loud sounds, bright colours, intense odours, huge beautiful buildings easily than smaller ones.
3. *Contrast, change, and variety:* Novelty and changes bring more attention to the individual than the routine occurrences. ***For example,*** we do not notice the ticking sound of a clock hung on the wall until it stops ticking.
4. ***Repetition of stimulus:*** The repetition of stimulus has a great impact on securing one's attention. We tend to ignore a single reminder or stimulus, but if we are reminded repeatedly many times, it captures our attention. ***For example,*** while giving health education to the community, the important aspects of education are often repeated to capture the attention of people in the community.
5. ***Movement of the stimulus:***
   * The moving stimulus catches our attention more quickly than a stimulus that does not move.
   * We are more sensitive to objects that move in our field of vision, e.g. Advertisers make use of this fact and try to catch the attention of people through moving electric lights.

II. **Internal or Subjective Factors:** The internal factor such as **interest, motives, mindset, and attitude** influence the individual to respond to the activities that fulfill their needs and desires.

Types of subjective factors are:

1. ***Interest:***
   * Interest is the main responsible factor to attend any activity, it is said to be **the mother of attention.**
   * We like to attend some activities in which we have more interest. Children like to play cricket because they are more interested in it.
2. ***Motives:*** Our basic needs include **thirst, hunger, sex, curiosity, fear** are some of the factors that influence our attention. ***For example,*** labour pays attention to their work in the hope to get their daily wages to fulfill their basic needs.
3. ***Mindset:*** Individual readiness to respond determines his attention. The Individual expects the stimulus or occurrence as per the schedule, thus, it influences him to pay the attention to complete the given task. ***For example,*** as per the timetable, students know that exams are held at end of the semester, so they pay attention to their studies.
4. ***Moods and attitudes:***
   * Our moods and attitude also have a great influence on our attention towards any activity. Our attention is poor, when we are in a disturbed or angry mood.
   * Similarly, our favourable and unfavourable attitude also influences our attention towards the objects.

Duration, Degree, and Alteration in Attention:

Attention is not limited to one object or at a point in time, it considers more than one object. Some people can grasp number of objects in a short period or other words by paying minimum attention. This ability is evaluated in means of the **span of attention,** which varies from one individual to another individual and even from situation to situation.

Span of Attention:

* Span means the number of objects or events one individual can attend at a given time.
* The span of attention is a **threshold to perceive at a glance at a given duration of exposure**: At a glance how many letters, digits one can see and reproduce.
* **The period** is the extent or limit of the ability of the individual to attend to objects with concentration. The length of time the reader spends with a lot of concentration on what he is reading, without distracting his mind on any other activities, called the **span of the time.**
* The span of attention varies with **age, physical, mental, and emotional conditions.**
* When an individual can attend to material for a maximum period of time is called **span of attention.**

This attention can be either visual or auditory:

* ***The span of Visual Attention:*** In this, researchers have exposed the individuals to a number of activities or materials, then examined the individual's visual attention power. The duration exposure time is rated from 1/100 to 1/5 of a second.
* ***Span of Auditory Attention:*** In this, people's auditory attention span is examined by checking the volume of auditory impressions perceived by the individuals at a single period.

Test of the span of attention:

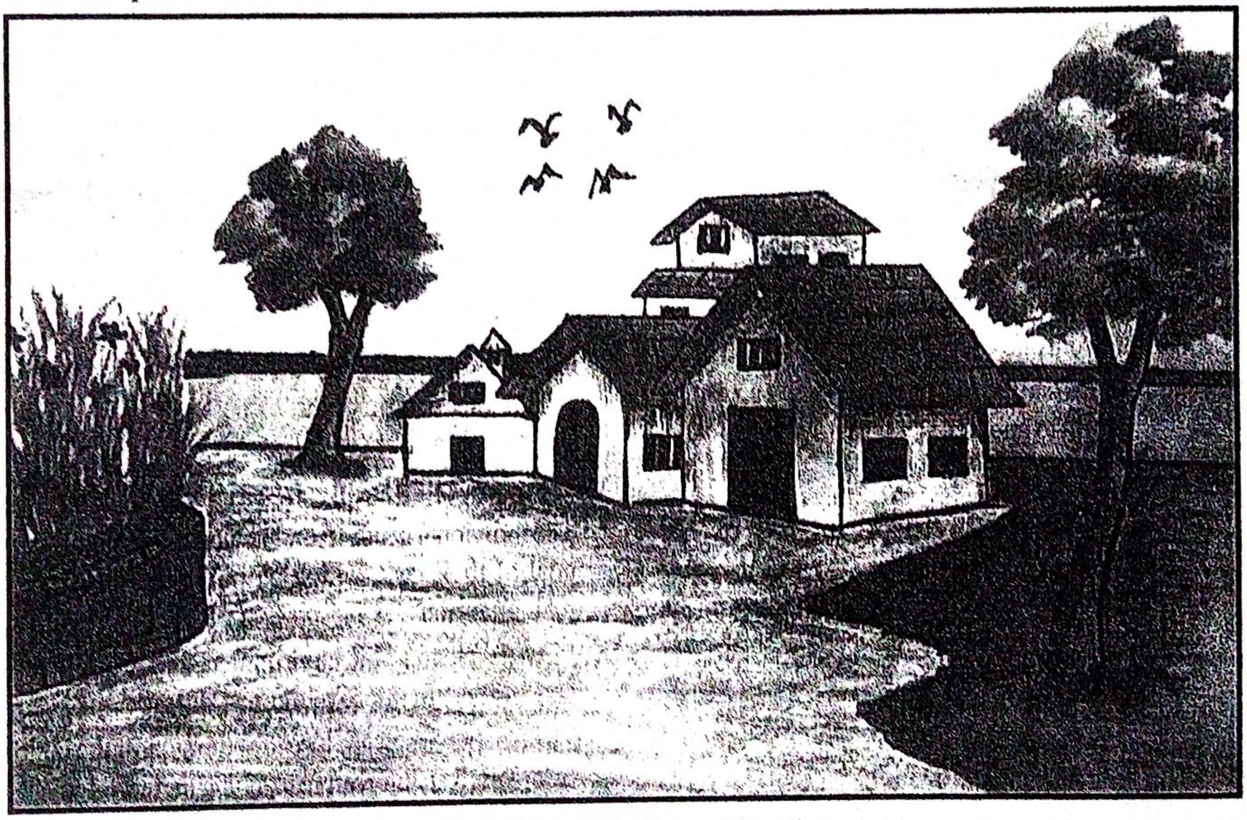

Fig. 6.2: Test of the Span of Attention.

To check the span of attention of the children, the examiner let the children see the diagram for a short period. Then the examiner can ask the **possible following question;**

* How many birds are flying in the given sceneries?
* How many trees are in the sceneries?
* How many doors does the house have?
* How many windows does the house have?
* Does the house have a chimney?

The span of attention is measured by Tachistoscope:

* A **tachistoscope** is a device that displays an image for a specific amount of time as we have mentioned above. It can be used to increase recognition speed, to show something too fast to be consciously recognized, or to test which elements of an image are memorable.
* Before computers became common, tachistoscopes were used extensively in psychological research to present visual stimuli for controlled durations.

Time Duration and Degree of Attention:

* It determines the individual's attention capacity to an object or activity without break.
* It is defined as the length of **one's ability to pay attention to perceived stimuli**.
* The important factors responsible witch are involved for the duration of attention.

The duration of attention is of three divisions;

1. ***Sustained Attention (Act of Fixation of MIND):***
   * In this attention, people **pay attention to objects with a strong desire**.
   * The people concentrate continuously on a single occurrence or object with a **will and specific objective**.
2. ***Shifting Attention:***
   * People can't concentrate and hold their attention for a long period, thus people prefer to shift their attention from one event to another periodically.
   * People like to shift their attention from one aspect of a situation to another.
3. ***Division of Attention:***
   * In this individual attend two or more activities at a single point of time.
   * The people pay attention to **more than one or two tasks at the same time**, people shift their attention from one work to another work quickly.

Alteration in Attention/Distraction:

* Due to some stimulus in the environment, people's attention is distracted from one activity to another activity.
* A distractor is a stimulus responsible to divert anyone's attention from one object to another. For eg. The children's mind is diverted from class in the classroom to the music played on the street.
* Distraction may be defined as any stimulus whose presence interferes with the process of attention or draws away attention from the object to which we wish to attend. *- H.R.Bhatia*
* A distraction may be defined as any factor which normally tends to break up attention. - *Prem Prakash*

Sources of Distraction:

The sources of distraction are two types

1. External factors.
2. Internal factors.

1. ***External Factors:***
   * It is also called **environmental factors**. These are commonly seen in our regular activities.
   * Noise, poor lighting, excessive music, poor ventilation, improper way of teaching, lack of AV aids in the classroom, destructive voice of the teacher are responsible external factors to distract our attention towards an object.
2. ***Internal Factors:*** Internal factors such as poor health, anger, anxiety, fatigue, inferiority, feeling of insecurity, boredom, less interest in the subject, etc can cause poor attention towards an object.

Types of Distraction:

1. ***Continuous Distraction:*** Here people are continuously disturbed due to noise made by street audios, noise at marketplaces or industries, etc. Adjustments to this kind of destruction are common.
2. ***Discontinuous Distraction:*** Here people are distracted periodically **at some sort of intervals.** Adjustments to this kind of destruction were found difficult.

Methods of Eliminating Distraction:

* The individual needs to understand the importance and goal of their task, which will help them to enhance their attention.
* Provide a favourable environment and situation to the task holders to improve their attention
* The teacher has to capture the attention of the children by creating an innovative teaching methodology in her teaching.
* The students or individuals need to be trained to improve their concentration.
* Attention distracting objects need to be removed from the surroundings.
* If the distraction occurs every day, people should be used to them and start ignoring them.

6.2. PERCEPTION - MEANING OF PERCEPTION, PRINCIPLES, FACTOR AFFECTING PERCEPTION

INTRODUCTION

Perception is a process by means of which people **are aware of their characteristics and those of their environment through the functioning of our sense organs.**

Perception is the process of **organization, identification, and interpretation** of sensory information in order to signify and understand the environment.

All perception includes **signals of the nervous system**, which results from physical or chemical stimulation of the sense organs.

It is not a passive acceptance of these signals and but is involved by **learning, memory, expectation, and attention.**

Perception comprises five senses, such as **touch, sight, taste, smell, and sound**.

Perception also involves the cognitive processes that need to process the information, such as recognizing the face of a friend or detecting a familiar object.

Meaning and Definitions:

* "Perception is defined as the way sensory information is **organized, interpreted, and consciously experienced**".
* "Perception is defined as our recognition and interpretation of sensory information".
* "Perception includes all processes by which an individual receives information about his environment—seeing, hearing, feeling, tasting and smelling". - ***Joseph Reitz***
* "Perception is the process of becoming aware of situations, of adding meaningful associations to sensations". - ***B. V. H. Gilmer***
* "The process of receiving, selecting, organizing, interpreting, checking, and reacting to sensory stimuli or data". - ***Uday Pareek***
* "The process by which individuals organize and interpret their sensory impressions in order to give meaning to their environments". - ***S. P. Robbins***

Importance of perception:

* Perception helps in molding the personality of an individual.
* Perception helps in organizing and interpreting the objects located around us.
* Perception plays an important role in the outcome of the behaviour of people because, we act based on what we see around us.

Principles of perception:

William James, American psychologist has stated that it will cause a big booming- buzzing confusion, if we understand the environment as it appears to us. We like to view the world as we want in a meaningful way.

In the process of perception, we pay attention to specific objects or stimuli and try to interpret them. Similarly, when we are exposed to discrete stimuli, we organize them in a form and perceive them meaningfully than they appear.

The Gestalt Psychologist has explained that the brain creates an articulated perceptual experience by perceiving a stimulus as a whole than perceiving discrete entities.

This is more meaningfully explained in the **principles of perception:**

1. **Figure-ground Relationship:** As per this principle, people are able to perceive the figures more meaningfully in a given context, figures cannot be separated from the context. For example, when we write words with a white chalk piece, the background of the blackboard is perceived easily.

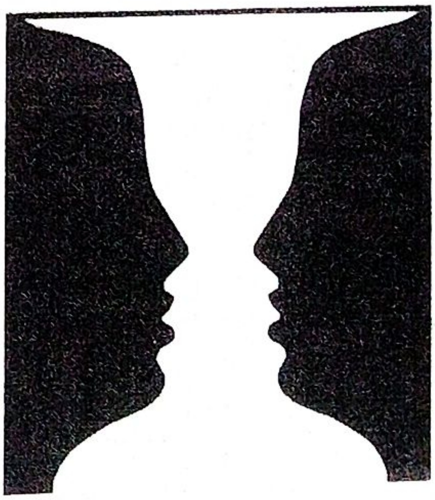

Fig 6.3: Reversible configurations.

In the above image, two faces can be seen in the background of white colour. And also white background can be perceived as a vessel in the background of two faces.

2. **Grouping of Stimuli in Perceptual Organisation:** According to this principle, the objects can be perceived meaningfully when they are grouped. They are as follows:

   a. ***Proximity:*** Proximity is referred to as closeness. The objects which are located near to each other can be perceived meaningfully by grouping them.

Fig 6.4: Proximity.

The stars in the figure show that each star is locating near to others and they are perceived together as groups or single figures.

   b. ***Similarity:*** In this, the various objects possess similar characteristics, they are perceived as one group.

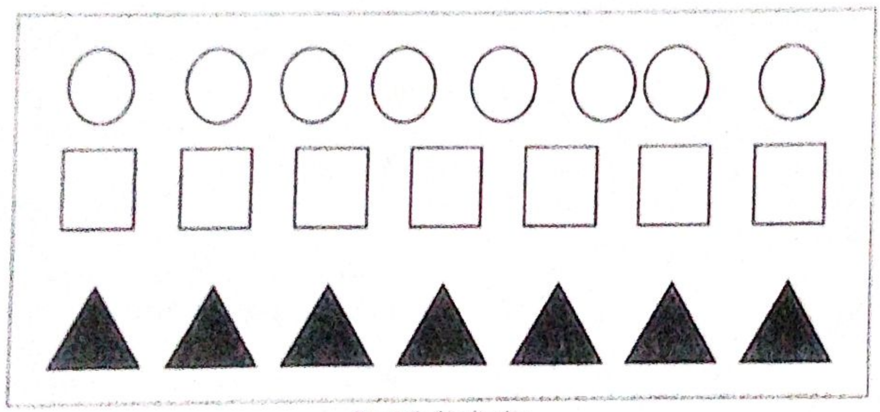

Fig 6.5: Similarity.

In the above figure, the objects are grouped as per similar characteristics of objects, i.e. all circles, squares, and triangles are grouped distinctly.

   c. ***Continuity:*** Any object or stimulus that expands over a similar direction or path will be perceived as a whole.

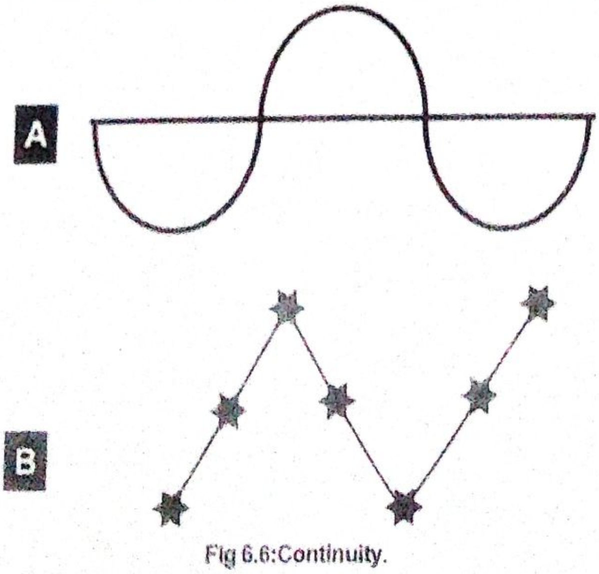

Fig 6.6:Continuity.

In the above (A) figure, though their breakage in curves, but the lines are continued to some extent. In (B), the dots are perceived as existing in the same line of direction continuously.

d. Closure: If the object is presented with gaps, we tend to perceive the image as a complete one by filling the gaps psychologically.

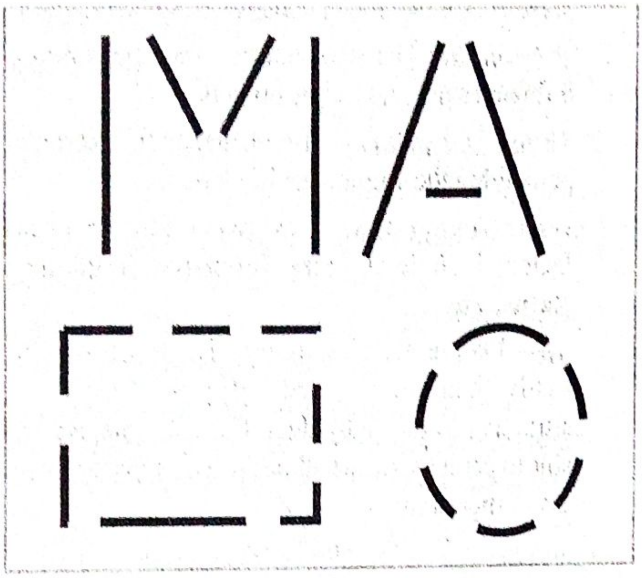

Fig 6.7: Closure

In the above image, the gaps are filled psychologically and perceived as letters M and A, circle, and rectangle.

e. Symmetry: The stimulus which is having symmetrical designs is perceived as groups.

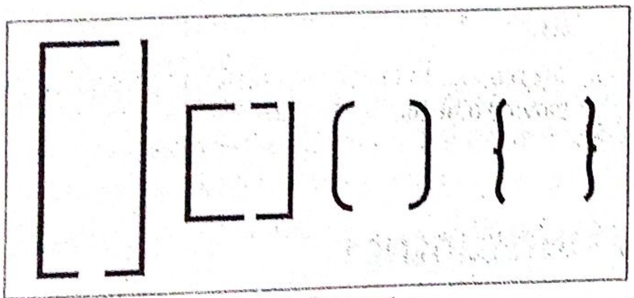

Fig 6.8: Symmetry.

In the above image, the brackets of different shapes are shown, perceived meaningfully, because they are grouped and perceived as brackets.

3. Perceptual Constancy: It refers to **consistency or stableness** in perception. People are inclined to perceive the stimulus relatively constant in unchanging shape and size, though there is some change in the stimulus that they receive.

Types of perceptual constancy:

*a. **Depth Perception:*** It checks the ability of a person to perceive the distance is known as **depth perception**. It is very important to judge the person's ability to perceive the distance between him and objects.

b. Cues: Depth perception becomes easy when we have certain cues. These cues are useful to estimate the **distance between one individual to another**.

The following are the types of cues;

i. Monocular cues: These cues function when people view objects with a single eye.

* ***Linear perspective:*** The distances separate the figure of the object seem to be smaller. For example, when we stand on the railway track, looking at the other end, it seems that the tracks would seem to run closer and closer together.
* ***Aerial perspective:*** The objects which locate near to us appear to be clear than the distant objects. For example, a hill far from distance appears farther away, because the details do not seem clear.
* ***Interposition:*** If any stimulus or object blocks the object, the front object seems to be nearer than the object covered by another one.
* ***Gradient structure:*** A gradient is a continuous change in something. Usually, the regions closer to the observer have a coarse texture and many details. As the distance increases, the texture becomes finer and finer. In the below figure, structures that are nearer appear larger than the distant one which appears smaller as they move away.

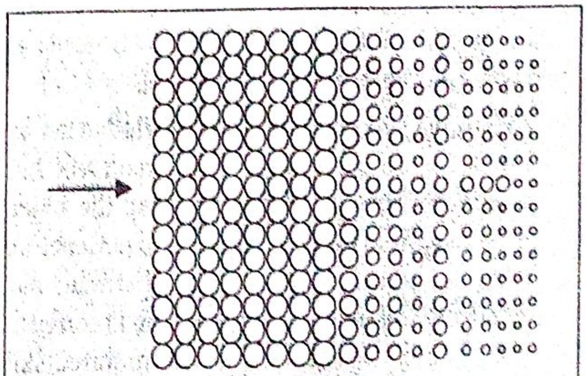

Fig 6.9: The structures appear longer when closer and smaller as they move away.

ii. ***Binocular cues:*** When we perceive the depth by closing both eyes called **binocular cues.**

* ***Retinal disparity:*** The image of the stimulus or object falls on both the retinas, which may differ. This difference is prominent when the object locates closer to the observer. Based on an association between the distance and the amount of disparity, the depth can be perceived.

* ***Convergence or divergence of eyeballs:*** The eyeball is converged when the objects move near our eyes. In contrast, the eyeballs will be diverged, if the object moves away from the eyes. This process functions as a binocular cue to perceive the depth.

4. **Perception of Movement:** When an object appears at different places at different timings, we perceive that the object is in motion. This process is known as the perception of the movement.

5. **Apparent motion:** Sometimes we perceive that the objects are moving, but in reality, these objects are static and not moving. ***For example,*** when we are travelling fast on a bus, the trees, plants, and other non-moving objects seem to move in the opposite direction.

Factors affecting perception:

There are an individual differences in perceptual abilities. Two people may perceive the same stimulus differently

The factors affecting the perception are two types:

1. Internal(endogeneous) factors.
2. External(exogeneous) factors.

1. **Internal Factors:** These factors include one's needs, desires, personality, and experience, etc.

a. ***Needs and motives:*** An individual's perception of the object is influenced by their own needs and desires during that particular period.

b. ***Experience:*** Experience and knowledge act as a foundation for perception. The person who has a good experience is able to perceive the objects differently from others. Adequate experience also helps the perceiver to understand stimuli more accurately. For example, A nurse who is competent in providing CPR can impart the required skills when a patient needs them.

c. ***Mental set:*** It is the readiness to receive some sensory input. Such anticipation keeps the people ready with good attention and concentration. For example, when we are expecting the arrival of the train, we listen to its horn carefully even if there is a lot of noise disturbance.

d. ***Cognitive styles:*** People are different in processing the information they receive. Every person has their skill in understanding the situation.

e. ***Personality:*** The personality of the person also influences perceiving the objects.

f. ***Values and attitude:*** Also influence the person in perceiving the stimulus or objects.

2. **External(exogeneous) factors:** The external factors include **the size, intensity, frequency, status,** etc.

a. ***Size:*** People are able to perceive larger objects easily than smaller ones. Big sizes are more attractive to the individual. For example, we are able to perceive the headlines of a newspaper easily than other content.

b. ***Intensity:*** People tend to perceive the stimulus, which has more intensity.

c. ***Frequency /Repetition:*** Recurrent stimuli have more perception than a single one.

d. ***Contrast:*** Contrast colours receive a quick and discrete perception by the individual.

e. ***Status:*** Researchers suggest that people with high status often exert more influence on the perception of an individual as compared to those holding low status.

f. ***Movement:*** People have a different perception of moving objects.

6.3 INTELLIGENCE

INTRODUCTION

Intelligence is one of the important characteristics possessed by individuals. **David Wechsler** stated that the intelligence of a person is the **capability to adjust to the world.** He had also mentioned that intelligence is the aggregate or global capacity of the **people to think logically, act meaningfully, and deal effectively with the environment.** It contains the **power of adaptation** of people to their milieu and their ability to **abstract the thinking process.**

6.3.3. CLASSIFICATION OF INTELLIGENCE

Howard Gardner's developmental psychologist in 1983 had explained nine types of intelligence.

1. Naturalist Intelligence:

* The people with this intelligence are able to **distinguish the living things such as plants and animals** and also they are very **compassionate to the natural world** such as clouds, rocks, etc.
* Naturalists love spending more time out of the home with the environment, plants, animals, etc.

2. Musical Intelligence:

* People with this intelligence can **differentiate pitch, rhythm, timbre, and tone.**
* This intelligence enables us to recognize, create, reproduce, and reflect on music, as demonstrated by composers, conductors, musicians, vocalists, and sensitive listeners.
* Musical intelligence is not a talent possessed by musicians or top singers. It is often featured by having a hobby of listening to the songs.

3. Logical-Mathematical Intelligence:

* Logical-mathematical intelligence helps people **to make calculations, consider propositions and hypotheses**, and carry out complete mathematical operations easily.
* This will able them to understand the relationship between different facts and develop symbolic thoughts. This intelligence is well developed by mathematicians, scientists, and detectives.

4. Existential Intelligence: This intelligence enables the individual **to tackle deep questions about human existence**, such as the meaning of life, why we die, and how did we get here.

5. Interpersonal intelligence:

* This intelligence **allows people to understand, and interact with others easily**. People with this intelligence possess effective verbal and nonverbal communication.
* It will enable us to understand the motives, desires, and needs of others by interpreting their facial expressions and body language.
* The individual who has these qualities become the leaders in their peer group.

6. **Bodily-Kinesthetic Intelligence:**

* This intelligence enables the individual to acquire a variety of **physical skills through the mind-body union.**
* This intelligence helps people to acquire motor abilities such as **athletics, dancing, hand-eye coordination, etc**.
* Athletes, dancers, surgeons, and craftspeople exhibit well-developed bodily-kinesthetic intelligence.

7. **Linguistic Intelligence:**

* People with this intelligence possess language skills such as writing and speaking. People can learn more languages.
* This intelligence is high poets, novelists, journalists, and effective public speakers.
* The people who possess this intelligence enjoy **writing, reading, telling stories,** etc.

8. **Intra-personal Intelligence:**

* Intra-personal intelligence will enable us to understand **their thoughts, feelings, weaknesses, needs, and desires.**
* **Self-awareness and self-concept** are the main constituents of this intelligence.
* This intelligence is high in psychologists, spiritual leaders, and philosophers.

9. **Spatial Intelligence:**

* It is one's ability to **manipulate the space, patterns as well as three-dimensional.**
* The abilities of this intelligence include **mental imagery, spatial reasoning, image manipulation, graphic and artistic skills, and active imagination**.
* Sailors, pilots, sculptors, painters, and architects all exhibit spatial intelligence.

INTELLIGENCE QUOTIENT

* IQ is the measure of intelligence of people
* IQ is a measure of relative intelligence that is determined by a standard test. These tests determine that majority of the children can reach the complexity, while some children are unable to reach or slow in reaching those same levels.
* A six years old child who cannot perform more than three years old child, this child's mental age is considered to be three years.
* **The Wechsler Adult Intelligence Scale (WAIS)** is a commonly used IQ test for adults.

IQ measurement:

* **Wilhelm Stern** divided the mental age by the chronological age to get a mental quotient.
* For example, for a six-year-old child who is able to perform only what a 3 years old child could perform, his mental quotient is 0.5.
* **Lewis Terman** later multiplied the mental age quotient by 100 to remove the fraction and the intelligence quotient (IQ) was born.
* IQ is the score you get on an intelligence test. Originally, it was a quotient(a ratio): IQ=MA/CAx100 (MA= Mental age, CA is chronological age).
* Mental age/Chronological agex100= Intelligence Quotient.
* The six-year-old with the mental Quotient of ½ has an IQ of 50
* The majority of people have an IQ between 85 and 115

Levels of IQ:

* Under 70 - mentally retard.
* 70-80 - Border retard.
* 80-90 - low Average.
* 90-110 - Average.
* 110-120 - High Average.
* 120-130 - Superior.
* Over 130 - Very superior.

B. Individual Tests:

There are mainly two individual tests used in the measurement of intelligence.

1. Stanford-Binet Intelligence Test (1916):

* This test is a new variety of Binet- Simon tests. This test was discovered by two French Psychologists named **Binet and Simon in 1905**. Again this test was modified by American Psychologist **Lewis Terman with Binet** at Stanford University to assess the Intelligence of the individual. Subsequently, it is named as **Stanford-Binet Intelligence Test**.
* This test is performed usually for the individual age group between 2-23 years.
* The total time needed to perform this test is 30 to 90 minutes.
* The formula or calculating:

$$I.Q. = \frac{Mental\ Age}{Chronological\ Age} \times 100$$

* The child whose age of 6 years, the child is expected to have the mental age of 6 years.
* I.Q. = (6/6) x 100 =100. This test is known as the Standard IQ test throughout the globe.

Following are the five factors measured by the Stanford-Binet:

* Solid rationale.
* Knowledge of an individual.
* Quantitative reasoning.
* Visual-Spatial Processing.
* Working on Memory.

Items of the Stanford-Binet scale are as follows:

a. ***Vocabulary:*** Individuals are asked to define the 11 words correctly from an arranged list of 45. The test starts with "Orange", "Straw", "Puddle" etc.

b. ***Block counting:*** Three-dimensional image is shown, then the individuals are asked to count the number of blocks in which some blocks are hidden.

c. ***Abstract Words:*** Individuals are asked to define the following words: disappointment, curiosity, sorrow, amazement.

d. ***Finding Reason:***
   * Provide rationale, why pupils are instructed to keep silent in school.
   * Why do most people like to own a car than a motorcycle.

e. ***Word Naming:*** The individual are instructed to name the words as many as possible within a minute.

f. ***Repeating digits:*** Repetition of the order in the sequence of a) 4-7-3-8-5-9 b) 5-2-9-7-4-6

2. ***Wechsler Scale:*** The Wechsler Bellevue Scale is one of the foremost famous intelligence scales that was discovered by **Dr. David Wechsler in the year 1938.** The main objective of this test is to measure the capacity of the individuals in adapting to a new environment and the ability to solve problems.

Following are three Wechsler Scales:

a. ***WAIS (Wechsler Adult Intelligence Test):*** It measures the cognitive performance of the individual's age group between 16-74 years.

b. ***WISC (Wechsler Intelligence Scale for Children) :*** This scale is used for children between the age group of 6 to 16 years.

c. ***WPPSI (Wechsler Preschool and Primary Scale of Intelligence):*** This scale is used for children between the ages group of 4-6 years.

According to the Wechsler test I.Q. = (Actual test score/Average score for norm group) x100

This score helps to measure how one person is different from others in the group.

Wechsler intelligence test contains 11 sub-sets:

a. 6 of them are related to **verbal intelligence** and

b. 5 of them are related to **performance intelligence.**

a. verbal intelligence:

The items of the verbal scale are:

i. ***Information:*** Individuals are asked to answer the following questions such as answering questions like "What is steam made of ?" or "what is a pepper?"

ii. ***Comprehension:*** What is your action when you have observed that someone has forgotten their mobile when he leaves his place in the hotel?

iii. ***Arithmetic:*** In this, children are asked to solve the maths problems at a particular time.

iv. ***Similarities:*** In this, people are instructed to describe how things are like? ***For e.g.*** What is the difference between lion and tiger?

v. ***Digit Span:*** People are directed to repeat the given numbers.

vi. ***Vocabulary:*** Individuals are asked to define particular words.

b. Performance Scale:

The items of the performance scale are:

i. ***Digit symbol:*** It involves a key consisting of the numbers 1-9, each paired with a unique, easy-to-draw symbol such as a " + " (or) " > ".

ii. ***Picture completion:*** The people are asked to recognize the missing portion in compound pictures.

iii. ***Block Design:*** People are instructed to arrange the blocks as per the colour or size.

iv. ***Picture arrangement:*** Individuals are asked to organize the pictures as per the story phrases given to them.

v. ***Object Assembly:*** Jumbled Jigsaw like pieces are assembled to make a picture.

6.4. LEARNING -DEFINITION OF LEARNING, TYPES OF LEARNING, FACTORS INFLUENCING LEARNING - LEARNING PROCESS, HABIT FORMATION

INTRODUCTION

The personality, habits, knowledge, attitude, interests, and character of people depended on the learning. One of the most common features of humans is that they can learn, it is the universal and important nature of mankind. **The learning process begins in childhood and continuous until death.** Learning is a responsible factor **to modify the behaviour of children.** Learning makes a man control his environment and be able to modify it as he desires.

People are interested to make attempts to develop adaptive competencies as per the requirements of a fluctuating environment.

Two things are important for learning;

1. The presence of a stimulus in the environment and
2. The innate dispositions like emotional and instinctual dispositions.

* Learning involves **changing the behaviour of the individual,** which can be better or worse. Behavioural change **depends on practice and experience.**

* Learning is growth and that is acquired by experience. Learning brings efficiency in adjustments as a result of practice, insight, observation, limitation, or conditioning.

6.4.2. TYPES OF LEARNING

Types of learning are;

1. Learning by trial and error.
2. Learning by imitation (OBSERVATIONAL LEARNING).
3. Learning by Cognition process.
4. Learning by conditioning.

1) Learning by Trial and Error:

* This theory was introduced by **Lee Thorndike (1874-1947)**, who is called a **Father of Educational Psychology**. He has conducted an experiment called **trial and error**.

* In this learning, an individual learn to perform the skills by exhibiting repeated behavior, through this experience, people learns to perform the behaviors which results in rewards and avoids the behaviour that results in punishment,

* This method can be used when the learner is completely ready to work and motivated to focus on goals.

Experiment: Thorndike has developed an elegant cage called a puzzle box in which a cat was placed and observed the leaning response, stepping on a small lever, which will allow the cat to unlock the door and get out. The cat was provided with food when it unlocks the door. Again the cat was placed inside the cage. After so many trials, the cat has learned how to open the door. After that, it used to walk calmly to the lever and used to unlock with its paw, strolled via the door, and used to eat the food.

Laws of Learning Edward.L.Thorndike :

a. ***Law of the effect:*** Any repetition is followed by a reward that means a positive result. If any response is unsuccessful, the behaviour will be stopped.

b. ***Law of Exercise:*** It depends on the relationship between the repetition and the strength of the stimulus-response bond. It is mainly based on the law of use and the law of disuse. The learning activities such as singing, writing, dancing, typing, drawing are learned by the practice over a long period.

c. ***Law of readiness:*** Learning takes place successfully, if the person is ready to learn. If the nurse is ready to work in the hospital, it gives her immense satisfaction.

2) Learning by Imitation (Observation) :

* As per this theory, learning occurs by keen observation or imitation of other's behaviour.
* The social model plays a keen role in observational learning i.e., family, parents, friends, teachers, and society.
* ***Modelling:*** When we try to imitate other's behaviour in the process of learning is called modelling.
* ***Social Learning:*** When the process of learning takes place from family, friends, teachers, society, etc., it is called social learning.
* ***Imitation:*** Pure imitation is blind copying of other's behaviour. Usually, students do this, while they copy the mannerisms of their favorite teachers.

Observational Learning is four types;

a) Attentional Process.

b) Retentional Process.

c) Motor Reproduction Process.

d) Motivation.

a) Attentional Process: In this process, people need to observe the models closely to imitate their behaviour.

b) Retentional process: In this process, people need to organize and retain the experiences observed from others.

c) Motor Reproduction Process: In this process, people try to reproduce the behaviours in their own life situations that they have acquired from others.

d) Motivation: The actual or imagined reward of imitated behavior determines whether the behavior will extinguish or not.

3. Learning by Cognition process:

a) Insight learning:

* This theory explains to us to understand the association of different parts of the problem rather than the test and error.
* In this cognitive process, we use the thought process of learning. Learning occurs by mental activity or thought is called learning by cognition.

The theory of learning by insight- Experiment:

Kohler Conducted an Experiment on Chimpanzees and concluded that an individual learns the behaviour by their own ability, not by the trial and error behaviours.

In his experiment, a smart chimpanzee was kept in a cage and it was restricted to reach a fruit that was placed near the bar of the cage. The Kohler has made the availability of sticks to the chimpanzee. But, that stick was very short. Then the chimpanzee has picked the short stick and gazed around the cage through the bar. With the help of a short stick, the chimpanzee has scratched another long stick which is lying outside the bar of the cage. Both sticks were tied by the chimpanzee and reached the fruit and ate.

Kohler has explained that chimpanzee has learned to solve his problem by putting several pieces together into a meaningful whole.

b) Sign Learning:

* According to Tolman, learning is a total process. It occurs by cognition.
* Cognition consists of **knowledge, thinking, planning, inference, and purpose.**
* The learner through his experience recognizes some cues or signs and the relationships with goals.
* According to this theory, people learn new behaviours by following some important signs and cues.
* People follow some sort of Mental map to learn new things this is called **sign learning theory**

4. Learning by Conditioning

Learning that occurs by association is called **conditioning**:

There are two types of conditioning

a. Classical Conditioning.

b. Operant Conditioning.

a. Classical conditioning:

* It was started in the initial year of the 20th century by **Ivan Pavlov**, a Russian psychologist who was the **Noble prize winner** for the research on digestion.
* Classical conditioning is based on the association of paring of an originally neutral stimulus with a response-producing stimulus. This pairing of stimuli eventually produces a condition or learns a response. **Stimulus-Response Produce Behaviour or Actions.**
* Classical conditioning could take place only when two events to be associated occurred close together.
* In classical conditioning, a neutral stimulus paired one or more times with a biologically significant stimulus acquires the power to elicit a behavioural response in the absence of the biologically significant stimulus.

Pavlov's Experiment:

In this experiment, the researcher first attached a capsule to the dog's salivary gland to measure salivary flow. A bell was rung, every time the dog Sam was given meat powder. This was repeated several times. Later the researcher was noticed that the dog started salivation at the mere sound of the bell, without the wheat powder being provided. The dog had been conditioned to respond to a new stimulus, which was a previously unconditioned response.

The meat powder is the unconditioned stimulus (UCS), the celebration is the unconditioned (CS) response, bell is the conditioned stimulus, and salivation at the sound of the bell is the conditioned response.

Pavlov's theory is that CS (bell) simply as a result of pairing with the unconditioned stimulus (meat powder) acquires the capacity to substitute for the unconditioned stimulus in evoking the response. This means that an association is formed between the conditioned stimulus and the unconditioned stimulus so that the conditioned stimulus, becomes the equal of the unconditioned stimulus in eliciting response.

He has explained that this association took place in the brain. Two areas of the brain are involved, one for the UCS and the other for the CS became activated during classical conditioning and the activation of the UCS area resulted in a reflex or automatic response.

b. Operant Conditioning

BF Skinner (1904 – 1990) explained that the best way to learn or understand the behavior, people need to look for the **causes of action and consequences**. For example, a child is learning to complete his studies or complete the homework because he gets some time to play or the teacher may punish him if he does not complete the homework.

The operant condition has three types of desirable and undesirable consequences that influence behaviour

* Positive reinforcement.
* Negative reinforcement.
* Punishment.

Positive Reinforcement:
The stimulus needs to be added to enhance the desirable behaviour Positive consequences; behaviour occurs more frequently (i.e. praise given).

Negative Reinforcement: Remove the stimulus something unpleasant, undesired is removed by behavior or does not happen at all

Punishment: Decrease the undesirable behaviour Consequence of behaviour is negative – Behaviour has been punished. Physical punishment used by society, parents, and others.

6.4.3. FACTORS INFLUENCING LEARNING

Three factors that influence learning are :

I. Factors associated with Learner.
II. Factors related to learning material.
III. Factors related to learning methods.

I. FACTORS ASSOCIATED WITH LEARNER:

The learner is the focal point in any learning. Without learners, there cannot be learning. The following are some of the factors associated with learners:

1. ***Motivation:*** It is the most important factor to motivate the learner for new learning. In addition to motivation, the learner should have a definite goal. It will direct the individual on the right path to achieve the goal.
2. ***Readiness and willpower:*** The learner has to be ready to learn the new behaviour. He must have strong willpower to overcome hurdles and problems. The behaviour of readiness helps an individual to develop a positive attitude.
3. ***The ability of the learner:*** This refers to the level of intelligence, creativity, aptitude, etc are required for learning. Intelligence helps the individual to learn and understand better.
4. ***Level of aspiration and achievement:*** Learning depends upon the level of aspiration of the individuals to achieve goals. If the aspiration level is too high, the learner will work hard and achieve more. However, the aspiration level should be as per the ability of the learner. Otherwise, it may affect negatively leading to feelings of inferiority.
5. ***Attention:*** People need to learn to concentrate their attention on learning. Attentiveness helps people to grasp learning material easily. Disturbance of attention affects learning.
6. ***General health condition of the learner:*** General health includes the physical and mental health of the learner. The learner should possess good physical health. Organic defects like blindness, myopia, hypermetropia, deafness, paralysis, autism, severe handicapped, etc., will affect learning. The problem in sense organs will lead to improper perception. Chronic illnesses may lead to fatigue and a lack of interest.
7. ***Fatigue:*** Muscular or sensory fatigue causes mental boredom and lethargy. Several factors in the home and school environment may cause physical and mental fatigue, such as lack of accommodation, bad seating arrangement, unhealthy clothing, inadequate ventilation, poor light, noise overcrowd, and poor nutrition.
8. ***Time of Learning:*** Morning and evening hours are the best periods of study. During the day, there is a decline in mental capacity. Experiments on

children have shown that there are great variations in learning efficiency during different hours of the day.

9. ***Food and Drink:*** Nutrition is responsible for efficient mental activity. Poor nutrition adversely affects learning. The type of food also has some effects. Alcoholic drinks, caffeine, tobacco, and such addictive items have an adverse effect on the neuromuscular system, and consequently upon the learning capacity.

10. ***Atmospheric conditions:*** Temperature and humidity lower mental efficiency. Low ventilation, lack of proper illumination, noise, and physical discomfort (as we find in factories and overcrowded schools) hamper the learning capacity. Distractions of all sorts affect the power of concentration and consequently the efficiency of learning.

11. ***Age:*** Learning capacity varies with age. Some subjects can better be learned at an early age, and some during adulthood. On the evidence of experiments conducted, Thorndike says that mental development does not stop at 16 or 18 but increases up to 23, and halts after 40. Learning proceeds rapidly between 18 and 20 remains stagnant till 25, and declines up to 35.

6.4.5. HABIT FORMATION

* A habit is a simple form of learning and a change of behaviour with experience.
* It is an automatic **response** to a specific situation that is attained by people normally as a result of repetition and learning.
* When a behaviour is developed to some extent that it is highly automatic, it is called **habit**. Generally, the habit does not require conscious attention.
* The term habit is strictly applicable only to motor responses, but often applied to habits of thought, perhaps more correctly termed attitudes.
* Habits play an important role in our daily life. All of us acquire different habits. They are part of our life.
* Habits may be good or bad. Hard-working, writing, reading, regular exercise, meditation, etc. are examples of good habits. Alcoholism, drug addiction, lethargy, procrastination, telling lies, dishonesty, stealing, deceiving others, escapism, etc. are the examples of bad habits.

Basis of Habit Formation:

* Habit formation may be explained in two terms. They are **Physiological and Psychological**.
* **The physiological basis** is related to our nervous system. According to this, when an act is repeated more times, a clear nervous connection is formed, leading to a pathway.
* **The psychological theories** explain that habits are acquired natures.
* According to these theories, any learning process or experience gained by an individual is retained.
* When this learning experience is repeated, it is firmly retained. This ability to retain helps us to get it strengthened and becomes a habit.

Types of Habits: Habits are divided into **three types** depending upon the nature of activities.

1. ***Motor habits:*** These habits refer to the muscular activities of an individual. These are the habits related to our physical actions such as standing, sitting, running, walking, doing exercise, maintaining particular postures of the body, etc.
2. ***Intellectual habits:*** These are the habits related to the psychological process requiring our intellectual abilities such as good observation, accurate perception, logical thinking, use of reasoning ability before taking decisions and testing conclusions, etc.
3. ***Habits of character:*** We express some of our characters in the form of habits.

For example, helping others who are in need, trusting people, being honest, talking in a friendly way, time management, hardworking, keeping our dress clean and tidy, etc.

These habits will have the essence of feelings and emotions; hence, these are also called **emotional habits.**

Measures for Effective Habit Formation:

The following are measures that help in habit formation.

a. Make a good start:

* A good beginning is considered to be is half done.
* We should have strong motivation and determination of the mind to start any task.
* We should not have confusion in our minds. For example, a nursing student decides to start to study at a fixed time for a fixed length of period.
* He or she should start as decided and should not hesitate on the first day itself.

b. Keep regular practice:

* It is essential to practice the new habit formation regularly until it becomes a routine in our life.
* Postponement or interruption should be avoided because it weakens our habit formation.
* For example, giving some excuses like headache, lack of interest or mood, and postpone the work- should be avoided.

c. Choose a favourable environment:

* Good habit formation depends upon the encouraging atmosphere also.
* For example, for a student who wants to work hard, there must be a company of hard-working students and not lazy fellows who have no interest in studies.

d. Do not stop until the goal is achieved:

* Once a habit is formed it is to be strengthened.
* Hence, it should be continued until it is firmly rooted.
* Meanwhile, we should enjoy the new habit, so that we find more interest to continue the practice. For this purpose, we may keep thinking of the positive effects of that new habit.
* For example, understanding the subject matter, scoring good marks, getting good results, achieving a good job, etc.

6.5 MEMORY-MEANING AND NATURE OF MEMORY, FACTORS INFLUENCING MEMORY, METHODS TO IMPROVE MEMORY, FORGETTING:

The memory comprises memorizing what has been learned. It is a mental activity that consists of three stages they are **learning, retention, recall, and recognition.**

For example, remembering a person's name, the name was learned by us at some previous time. It was written in the mind when we did not think about it. When we needed this name again, many similar names came to our mind before we get the actual name, the one we needed was finally recalled or reproduced and recognized.

These three processes of memory are the most important in the ability to recall or reproduce. Thus, experts find the memory is one's ability to recall past experiences.

Good memory depends on effective learning, which ensures retention, for which intelligence helps in recalling the right thing at right time.

Remembering becomes quick if the learning material is studied and clearly understood. The significant material is more easily understood than meaningless material.

Forgetting means a fact or group of ideas are failed to recollect is called **forgetting.**

We tend to forget the facts which we do not consequently make use of it.

We failed to recollect the things due to a **lack of practice and repetition.**

Memory is the ability to take information, store it, and recall it at a later time. In psychology, memory is broken into three stages: **encoding, storage, and retrieval.**

The Memory Process:

1. ***Encoding (or registration):*** The process of memory comprises **receiving, processing, and relating the data.** In the form of chemical and physical stimuli, encoding is the process of passing information from outside the environment to the senses.
2. ***Storage:*** It is the development of a permanent record of encoded information. Storage is the second memory stage in which we store the data for longer periods.
3. ***Retrieval (or recall, or recognition):*** In this phase, the stored data is recalled and used whenever it is required by the process of brain activity. The third process is the retrieval of data that we have stored.

Types of Memory:

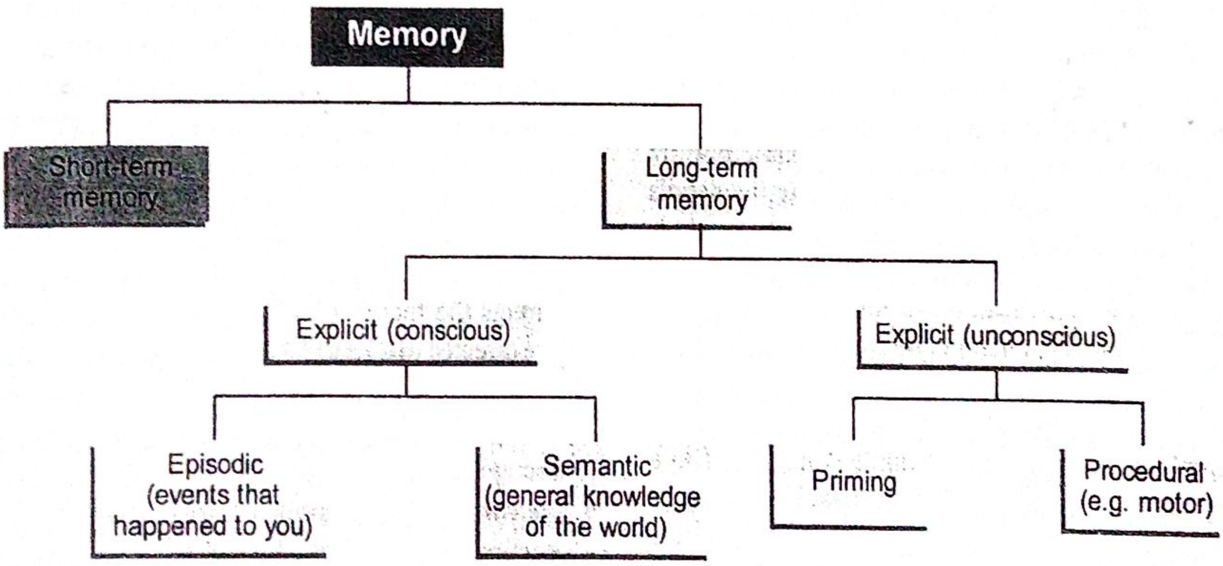

Fig 6.12: Types of memory.

1. Short-Term Memory:

* Short-term memory is also called **working memory**. It stores only a few things and lasts for about 30 seconds. The information is stored in our brains for a short period of time.
* The retention of data in the brain is longer than the immediate memory. This is also called a **recent memory.**
* However, short-term memory can be shifted to long-term memory by using rehearsal techniques.
* For example when someone gives you a phone number verbally and you say it to yourself repeatedly until you can write it down. If someone interrupts your rehearsal by asking a question, you can easily forget the number, since it is only being held in your short-term memory.

2. Long-Term Memory

In long-term memory, people are able to store the data for a longer period. Long-term memory has the capacity to hold huge information, some memories can be stored until we die.

This memory is also referred to as permanent memory or remote memory. In this people have a better span of attention than other types.

The long term memory has two types, they are:

a) Explicit memory.
b) Implicit memory.

a) Explicit Memory

* It is also called **conscious memory**, this is used to recall the information whenever it is required.
* For example, whenever you fill the college applications, you must consciously recall your full address, telephone number, and social security number.

The explicit memory is subdivided into two types as they are i) Episodic ii) Semantic.

i. Episodic Memory

* It is a form of declarative memory; people with this memory have the ability to remember first-hand experiences from their life.
* This type of memory is associated with some events and experiences of life. For example, explaining the accident. Observation of any event in our life is stored easily and it helps to narrate the details almost through life.

ii. Semantic memory: This type of memory helps an individual to recall the relationship between the events that happened in one's life.

b). Implicit Memory: Implicit memory is also called "**unconscious memory**". It is the process of retaining information from a moment that has happened in a time that cannot be specifically recalled.

It is two types;

i. Procedural memory :

* Individuals are capable to retain the data for a long period, if they would learn something in form of procedures. It is a form of implicit memory.
* It is useful in learning skill-based information, it remains permanently until the person dies.

ii. Priming Memory:

* It is an unconscious form of human memory related to the perceptual identification of words and objects.
* Try to remember the following words in sequence: College, Students, girlfriend, birthday, party, Pizza hut, Secretariat, father, escaped, inquiry, and sorry.

Factors Influencing Memory:

Our memory is certainly one of our most interesting talents. It allows us to store information, to reason, to understand, and of course to learn. Certain factors can influence the functioning of this highly complex brain function.

There are two factors responsible to influence our memory;

1. *Meaningfulness of the material to be memorized:* The data which is useful, meaningful something that suits, needs, or motivates can be learned and memorize easily. Eg. The words such as the rat, cap, tent, etc. can be learned quicker than the words such as NAL, SOK, PAB as they do not have meaning.
2. *Amount of the material to be memorized:* The size, time, quantity of the data can also affect the process of memorization.

Methods to Improve Memory:

Methods of Memorizing;

1. *Recitation Method:* The learner tries to recite and recall the content after reading once or twice. It is useful in self-appraisal and self-evaluation and then the learner can write down the facts that he is unable to recall.
2. *Whole and Part Method:* The learner is trying to remember the facts from beginning to end one stretch by reading the whole or part by part.
3. *Spaced and Unspaced method:* This type of memory directs the theory of work and sleep. The students are directed to take a rest after memorizing something.
4. *Recognition:* It is a process that occurs in thinking when some event, process, pattern, or object recognition occurs.
5. *Relearning:* It is another form of remembering, it occurs by relearning. Relearned information may return quickly, even if it has not been used for many years.

Tips for memory improvement:

* *Brain Exercise:* The more you keep the brain active, the better you will be able to process and remember the information.
* *Aerobics:* The best way to improve our memories seems to increase the supply of oxygen to the brain, which can be done by aerobic exercises.
* Pay attention to objects carefully.
* Involve as many as senses possible.

Unit-6 :: Cognitive Process

* Relate information about what you already know.
* Organize the information.
* Understand and be able to interpret complex material.
* Rehearse the information frequently and overlearn.
* Be motivated and keep a positive attitude.
* Have adequate rest and sleep.
* Manage stress.
* Avoid smoking.

6.6. THINKING- TYPES, LEVEL, REASONING, AND PROBLEM SOLVING.

INTRODUCTION

Thinking is a mental process that deals with thought and ideas.

Thinking is a cognitive activity and it directs us to achieve some end or purpose. It is used when the problems need to be solved. In **thinking mind is involved more** than motor activity.

Psychologists have defined thinking as a mindful activity, which involves many parts of our body and mind receptors, connectors, and effectors i.e. sense organs, nerves, and muscles. Out of these connectors, the neural mechanisms in the brain are the more important factor. When the thinking process is intense, we notice tremors in our bodies. Some people clench their hands when they cannot solve their problems easily.

The nurse needs to study the thinking process, reasoning, and problem-solving as she has to think critically when she delivers nursing care to her patients. For every nursing action, she has to give appropriate reasoning.

Definitions of Thinking:

* "Thinking is a mental activity in its cognitive aspect of mental activity concerning psychological objects ".
  - *Ross*
* "Thinking is behaviour which is often implicit and hidden and in which symbols (Images, Ideas, Concepts) or ordinarily employed." - *Garret*
* "Thinking is processing the information mentally or cognitively by rearranging the information from the environment and the symbols are stored in the memory of his/her past."
* "Thinking is a symbolic representation of some event to train ideas in a precise and careful that began with the problem."
* "Thinking is a mental representation newly formed through transformation of information by interaction, attributes such as the assessment of mental abstraction, logic, imagination, and problem-solving."

Types of thinking:

It is difficult to classify all types of thinking, there are so many individual differences between one thinker and another.

Following are the types of thinking;

1. Perceptual or Concrete Thinking.
2. Conceptual or Abstract Thinking.
3. Reflective Thinking.
4. Creative Thinking.
5. Non-Directed or Associative Thinking.
6. Free Thinking.
7. Controlled Thinking.
8. Critical Thinking.

1. ***Perceptual or Concrete Thinking:*** This type of thinking is simple. The base of this kind of thinking is a perception that is understanding of conciseness according to one's experience.

2. ***Conceptual or Abstract Thinking:*** In this, people use the concepts and generalize the ideas. This type of thinking is superior to concrete thinking as it helps to save time and effort in resolving issues.

3. ***Reflective Thinking:*** It is a higher form of thinking. The following are the aims of this type of thinking:-

* It solves difficult problems rather than simple problems.
* It requires the organization of related experiences and finding new solutions for the problems.
* In this type of thinking, mental activity does not undergo any mechanical trial and error type of effort.
* In this type of thinking an intellectual thoughtful approach is involved.
* All facts are arranged in a logical order to get a solution to solve the issues.

4. ***Creative Thinking:*** It is primarily involved to generate something novel. The thinking scientists and discoverers are examples.

5. ***Non-Directed or Associative Thinking:*** This thinking is mainly associated with reasoning and procedures of problem-solving.

6. ***Free Thinking:***

* In free-thinking, the thought process is allowed much greater freedom of action.
* There are no reactions to reality or in time and space.
* There is no desire on part of the thinker to achieve a realistic goal.
* The process of imagination, daydreaming, nd dreaming is typical examples of thinking.

7. *Controlled Thinking:*

* In controlled thinking, we regulate and control the process of thinking.
* We see that our thoughts keep in close touch with reality and are directed towards the achievement of specific goals.
* Reasoning, problem-solving and creative thinking are examples of control thinking.

8. *Critical Thinking:* It helps an individual to step aside from his personal beliefs, prejudices, and opinions to clear out and discover the truth.

The Levels of Thinking:

Levels of thinking are as follows;

1. *Level-1 or Knowledge level:*

* It is a basic level that is acquired by thinkers in parts. Most thinkers are observers, they try to interpret the information that they see superficially. There is no reasoning involved at this level. This level of thinking is possible for most of us.
* They gather the information in facts, statistics but, they never question the rationale behind them.
* For example, the children are asked to identify and describe objects to interpret them. Thinkers are asked to recall from memory specific facts and figures.

2. *Level 2 or Comprehending and confirming:*

* At this level, thinkers understand the meaning of knowledge and then they conclude. People interpret the information with proper associations and meanings.
* This level of thinking consumes more time.
* Thinkers interpret and analyze the information in pieces they have observed. Then, they gather the whole information and to form a meaning.
* Level two thinkers synthesize better, build up, or connect separate pieces of information to form a larger, more coherent pattern.
* They are better at reorganizing or rearranging ideas to produce a more comprehensive understanding of the "big picture".

3. *Level 3 or Applying :*

* Attempting to try and understand the learned knowledge in different similar practical situations.
* This is also called **the alpha stage thinking**. These thinkers use knowledge in situations that they have learned. Nurses learn the procedures theoretically in the classroom, that knowledge is applied to patients in the hospitals.
* Thinkers at this level try to apply the learned knowledge in real-life situations.
* Thinkers at this level can perceive the information from a variety of viewpoints, standpoints, or positions to gain a more comprehensive and holistic understanding.

4. *Level 4 or Analysing:*

* At this level thinkers understand the whole in terms of its various parts.
* The whole information is breaking into parts, then they use the logic, cause and effect relationship, relating and distinct, distinguishing the design.
* You can motivate the children to use this kind of thinking by asking them to use the specific kind of material in the classroom project.
* Where is the interconnectedness between us, and how can we best support each other's as growing, learning human beings?
* It turns out that our greatest adversities and most complex problems in life are best overcome when we look at them with the next level of thinking.

5. *Level-5 or Synthesizing:*

* At this level, the thinker keeps the parts together form as a whole.
* We often observe the facts in parts put together to shape something new, suggesting solutions, and create new information.
* It includes the formation of theories and tests.
* Asking children how to use an array of materials to create something new.

6. *Level 6 or Evaluating :*

* It is the highest level of thinking.
* It entails making comparisons and judgments.
* It is the ability to evaluate and judgments about a particular idea, its application, and suitability.
* A person at this level defends his work effectively.

6.7. APTITUDE- CONCEPT, TYPES, INDIVIDUAL DIFFERENCES, AND VARIABILITY

CONCEPT OF APTITUDE

* An aptitude is a constituent of the capability to perform a certain task at a certain level.
* The outstanding aptitude of people is considered as a "**talent or passion**".
* An aptitude involves the physical or mental process.
* Aptitude is the innate potential to perform certain tasks whether it is developed or undeveloped.
* Aptitude is the ability to develop knowledge, understanding, learned, or acquired abilities (skills) or attitudes.
* The innate nature of aptitude is in contrast to possessing skills, knowledge, and achievements.
* Individuals are diverse from one another within themselves in the activities includes leadership, music, art, mechanical work, teaching, etc.

Certain aspects are inborn. For example Musical aptitude: people possess musical throat, working man has long and strong hands. It is also impacted by the environment, in which people are brought up. Both environment and heredity have an equivalent role in possessing aptitude.

* Ability: It explains the individual's ability in possession of a particular capacity or ability.
* Achievement: It describes what one has learned or acquired.
* Aptitude: It predicts future success in a relevant field.

Definitions of Aptitude:

* **Aptitude** refers to qualities characterizing a person's way of behavior which serves to indicate how well he can learn to meet and solve certain specified kinds of problems. - ***Bingham***
* It is a specific capacity or special ability indicative of one's probable success in a relevant field after getting the proper opportunity for learning or training.
* Aptitude is an intelligence that is associated with **knowledge, cognition, and other aspects**. For a successful life, job, training, or courses of study, we require aptitude (special abilities) rather than intelligence or general ability.

Types of Aptitude:

1. Sensory Aptitude.
2. Mechanical Aptitude.
3. Artistic Aptitude.
4. Professional Aptitude.
5. Scholastic Aptitude.

1. ***Sensory aptitude:*** These aptitudes are associated with the **sensory capacities and abilities** of the individuals. It comprises a sense of **hearing, sight, smell, taste, and touch**. In this kind, an individual can direct themselves and become in those areas of life where these can apply perfectly.
2. ***Mechanical Aptitude:*** Some people have a particular mind for the tasks which are associated to use mechanical skills and it determines the ability of their mechanical capacity. In other cases, a person may be good at mending vehicles, engines, machines, etc. Some people are good at using classy machines and make innovations in instrumentation.
3. ***Artistic Aptitude:*** Some people have artistic abilities. It has an aesthetic sense, It is the sense of what is beautiful. It is primarily a personal sense but may be enhanced by another's input or awareness. There are many categories of artistic aptitude: Musical, Dance, Graphic art, photography, poetry, acting, debate, writing, designing, etc.
4. ***Professional Aptitude:*** It is related to profession and occupation. It predicts the future success of the person. It includes Clerical, Legal, Teaching, Pilot, Navigation, Banking, Military, etc.
5. ***Scholastic Aptitude:*** This is academic in nature. It drives the learning process. It includes scientific, engineering, medical, commercial, sports, linguistic, etc.

Tests of Aptitude:

Aptitude tests are ;

1. Mechanical aptitude tests.
2. Clerical Aptitude Tests.
3. Musical aptitude tests.
4. Graphic Art judgment tests.
5. Professional and Scholastic aptitude tests.

1. Mechanical Aptitude Tests:

It comprises **sensory, motor, perception, spatial capacities**. Some of the common tests are;

* Minnesota Mechanical Assembly test,
* SRA Mechanical Aptitude test.

These tests contain questions of the following nature;

* Individuals are asked to put the parts of machines together.
* People are asked about the basic information and knowledge about mechanical tools.
* Individuals are asked questions about mechanical principles.

2. Clerical Aptitude Tests. It measures numerous specific abilities.

Some of the tests are **Minnesota vocational tests for clerical workers and Clerical Aptitude Test.**

These tests measure the following abilities;

a. **Perceptual Ability** of an individual.
b. **Motor Ability**: Speed and perfection of the use of a computer, duplicating machine, and typewriter.
c. **Intellectual Ability**: The ability to grasp the meaning of words and symbols quickly.

3. Musical Aptitude Tests: These tests check the musical talent.

It checks the following aspects;

* Discrimination of pitch,
* The intensity of loudness.
* The judgment of rhythm.

4. Graphic Art Aptitude Tests: These tests are developed to check the performance of talent for graphic art.

The major two tests are;

a. ***The Meier Art Judgment Test:*** In this, there are 100 pairs of representative pictures present in black and white. One piece is the masterpiece and the other one is a distortion of the masterpiece. The subjects are asked to recognize the more attractive, artistic, and more satisfying piece.

b. ***Horne Art Aptitude Inventory:*** In these subjects are asked to produce sketches from the given patterns of lines and figures. They created figures that are then evaluated to the reference standard.

The tests are;

* The Meier Art Judgment Test.
* Horne Art Aptitude Inventory:

5. **Scholastic And Professional Aptitude Tests.**

This checks the children's ability or interest in the studies of specific courses or professions such as engineering, medicine, law, business management, teaching, etc.

Some of them are:

a. *Stanford Scientific Aptitude test:* This test is used to measure the academic knowledge of the students.

b. Science Aptitude Test.

c. Teaching Aptitude Test (M.P).